

A Kuhnian Historiography of Computational Neuroscience

Executive Summary

This report develops a historiographic strategy for computational neuroscience using Thomas Kuhn's framework from *The Structure of Scientific Revolutions* (SSR). SSR treats the history of science as evidential for understanding how knowledge is gained, accepted, and assimilated, and it emphasizes that scientific development is structured by community-bound commitments rather than by a purely cumulative "inventory of facts." ¹

Within SSR, **paradigms** are "universally recognized scientific achievements" that supply *model problems and solutions* for a community, enabling **normal science** as paradigm-guided puzzle-solving conducted under shared standards. ² In this report, Kuhnian categories (paradigm, normal science, anomaly, crisis, revolution, incommensurability) are operationalized as historically tractable indicators: exemplars (canonical models and "worked" solutions), training and textbooks, conference/journal epistemic norms, and episodes where persistent anomalies (or technology-driven discontinuities) precipitate reorientation of core standards and problems. ³

A chronological reconstruction from early neuron modeling and cybernetics to contemporary big-data neurotechnology suggests that computational neuroscience often developed as a **problem- and method-defined** enterprise spanning multiple levels (molecules → cells → circuits → systems → behavior). A classic 1988 synthesis explicitly framed the field's aim as explaining how physical signals represent and process information, while also noting the absence of "successful large-scale theories" and describing the field as defined by the problems it seeks and the methods it uses. ⁴ This pluralism and multi-levelness complicate a strict "one field, one paradigm" reading of Kuhn's model, but they align with Kuhn's allowance that some paradigm changes affect subspecialties and that negotiated translation is central when communities diverge. ⁵

The report identifies several **candidate paradigms** (better: partially overlapping Kuhnian paradigms or Lakatosian "research programmes"): (i) conductance-based mechanistic modeling and simulation (e.g., HH-style modeling and compartmental dendritic theory), (ii) networks/learning and neural computation (connectionism and later deep learning), (iii) neural coding and information-theoretic representation, (iv) normative probabilistic inference and predictive processing, (v) data-intensive statistical/ML neuroscience grounded in large-scale recordings and standardized data infrastructures, and (vi) large-scale simulation neuroscience and "digital brain" reconstruction. ⁶

Kuhn's framework fits computational neuroscience most strongly when applied to **subcommunities** organized around shared exemplars, standards, and training pipelines (e.g., conductance-based modeling + simulation environments; high-density recording + latent-variable/statistical modeling). It fits less well as a total-field model because computational neuroscience is structurally interdisciplinary and has repeatedly expanded through instrumentation, data infrastructures, and institutional programs rather than by field-

wide replacement of a single dominant paradigm. These limitations motivate complementary historiographic lenses: Imre Lakatos ⁷ ' "research programmes" (coexisting hard cores and progressive/degenerative shifts), Larry Laudan ⁸ ' research traditions, Peter Galison ⁹ ' "trading zones" for cross-paradigm coordination, Ian Hacking ¹⁰ ' styles of reasoning, and Pierre Bourdieu ¹¹ ' field theory for institutional power and capital. ¹²

Kuhn's Framework and Its Scholarly Reception

Kuhn's core concepts with precise anchors to SSR

Paradigm (as exemplar-achievement). SSR defines paradigms as past scientific achievements that are both unprecedented (attracting adherents) and open-ended (leaving problems to solve), thereby defining "legitimate problems and methods." ¹ Kuhn also characterizes paradigms as "universally recognized scientific achievements" providing "model problems and solutions." ¹

Normal science. Kuhn's canonical definition: "normal science" is "research firmly based upon one or more past scientific achievements" acknowledged by a community as the foundation for further practice. ¹ Normal science is structurally **puzzle-solving**: puzzles test ingenuity under paradigm-given constraints rather than aiming to refute fundamentals. ¹³

Anomaly → crisis. Kuhn emphasizes that normal science "repeatedly goes astray" and that anomalies can resist alignment with professional expectation; when anomalies can no longer be evaded and "subvert the existing tradition of scientific practice," extraordinary inquiry begins. ¹

Revolution. In SSR, scientific revolutions are "extraordinary episodes" marked by a "shift of professional commitments," a tradition-shattering complement to normal science. ¹ Revolutions entail standards- and world-view shifts (often analogized to gestalt switches), and they produce outcomes "incompatible" and sometimes "incommensurable" with what came before. ¹⁴

Incommensurability and translation. SSR uses "incommensurable" to describe divergent ways of seeing and practicing science, including cases where there is "no common body of belief" and where parties may be "members of different language communities." ¹⁵ Kuhn's Postscript develops **translation** as a tool for cross-community comparison and persuasion, while insisting that translation is not equivalent to "going native" (internalizing a new conceptual scheme). ¹⁶ This provides a concrete historiographic handle: one can track where communication shifts from shared vocabulary to translation work (e.g., imported formalisms, new data standards, or algorithmic languages).

Disciplinary matrix and exemplars. In the Postscript, Kuhn refines "paradigm" by proposing "disciplinary matrix" as the constellation of symbolic generalizations, models, values, and shared exemplars that bind a community's practice. ¹⁷ This is especially useful for computational neuroscience, where "symbolic generalizations" (equations, objective functions), "models" (mechanistic vs normative), and "exemplars" (canonical datasets and benchmarks) can be identified in textbooks, software libraries, and conference norms.

Major Kuhn scholarship relevant to applying SSR to computational neuroscience

Kuhn's "paradigm" was rapidly recognized as polysemous. Margaret Masterman ¹⁸ famously counted "not less than twenty-one different senses" of "paradigm" in SSR and grouped them into broader families, pressing for analytical clarity about what "paradigm" denotes in practice. ¹⁹ This matters directly for computational neuroscience because the field's "paradigms" may be methodological (e.g., statistical latent-variable modeling) rather than metaphysical.

Kuhn's 1965/1970 debates with critics (in *Criticism and the Growth of Knowledge*) foreground methodological questions about normal science, theory choice, and incommensurability—issues that recur when comparing, for example, mechanistic simulation programs with normative Bayesian approaches. ²⁰

Later Kuhn scholarship often treats incommensurability in semantic/taxonomic terms rather than as wholesale irrationality. The Stanford Encyclopedia of Philosophy (SEP) highlights Kuhn's uses of "no common measure" and tracks how incommensurability relates to theory change and translation. ²¹ Paul Hoyningen-Huene ²² 's book-length reconstruction is a standard reference for Kuhn's philosophical development and is particularly helpful for distinguishing SSR's early claims from Kuhn's later refinements.

²³

A practical Kuhnian template for writing a field historiography

A Kuhnian historiography of computational neuroscience should treat "paradigm" not as a label for any popular idea, but as a historically anchored package of: (i) **exemplars** (canonical model solutions, benchmark tasks, reference datasets), (ii) **training mechanisms** (graduate programs, textbooks, toolchains, summer schools), and (iii) **community standards** (what counts as explanation, acceptable idealizations, validation norms). ²⁴ The historiographer then narrates (a) periods of normal science (puzzle-solving within a stable exemplar set), (b) anomalies (persistent mismatches or methodological breakdowns), and (c) reconfigurations where new exemplars and standards reorganize research practice. ²⁵

Origins and Chronological Development of Computational Neuroscience

Computational neuroscience's roots are older than the name: early neuron models predate modern biophysics, while mid-20th-century cybernetics and neural-network logic forged bridges between nervous systems and computation. ²⁶ By the late 1980s, influential syntheses defined the field's aim as explaining neural information processing and stressed that advances in experimental neuroscience, computing power, and simplified network models were jointly enabling more realistic simulations and multi-level bridging.

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Timeline table of milestones

The table below is designed as an **evidence-first scaffold** for a Kuhnian historiography: each milestone is tied to a primary or official source so that historiographic claims about "exemplars," "normal science," and "reorientations" remain traceable. ²⁸

Date	Milestone	Primary / official source(s) to cite	Why this matters for a Kuhnian historiography
1907	Louis Lapicque ²⁹ introduces an early integrate-and-fire neuron model	Abbott's historical note anchoring Lapicque (1907) and the integrate-and-fire lineage ³⁰	Establishes an enduring <i>exemplar</i> of idealization: neuron-as-threshold device, recurring across later paradigms (spiking models, inference models).
1943	Warren S. McCulloch ³¹ and Walter Pitts ³² publish a logical calculus model of neural activity	Original paper PDF copies ³³	Seeds the “neuron-as-logical/computational element” lineage—a conceptual ancestor for connectionism and ML-style paradigms.
1948	Norbert Wiener ³⁴ publishes <i>Cybernetics</i> (feedback, control, communication), later foundational for brain-as-information-processing framings	MIT Press ³⁵ notes first publication in 1948 ³⁶	Provides a high-level <i>disciplinary matrix</i> vocabulary (feedback, signals) that can unify heterogeneous modeling traditions. ³⁷
1949	Donald O. Hebb ³⁸ publishes <i>The Organization of Behavior</i>	Full-text copy and bibliographic anchors ³⁹	Stabilizes learning-as-synaptic-change as a guiding commitment later re-expressed in neural networks, plasticity rules, and RL.
1952	Alan L. Hodgkin ⁴⁰ and Andrew F. Huxley ⁴¹ publish the conductance-based HH model of action potentials	Open-access HH paper ⁴²	Canonical mechanistic exemplar enabling decades of “normal science” in conductance-based modeling and simulation. ⁴³
1961	Horace B. Barlow ⁴⁴ articulates principles for transforming sensory messages (efficient coding lineage)	Primary chapter PDF ⁴⁵	Helps establish a normative/information-theoretic tradition: explanation via optimality/information constraints rather than detailed mechanism.
1964	Wilfrid Rall ⁴⁶ 's dendritic-tree theory (input-output relations) becomes a foundational formalism for compartmental modeling	MIT Press OA chapter ⁴⁷	Provides a mechanistic and mathematical <i>exemplar</i> for mapping morphology → function, central to simulation neuroscience.

Date	Milestone	Primary / official source(s) to cite	Why this matters for a Kuhnian historiography
1976	Erwin Neher ⁴⁸ and Bert Sakmann ⁴⁹ record single-channel currents (patch-clamp era)	PubMed record ⁵⁰	Technology-driven expansion of what counts as “data” for mechanistic models (ion channels, parameters), often stabilizing normal science.
1982	David Marr ⁵¹ publishes <i>Vision</i> (levels of analysis; computational investigation)	Accessible PDF and publisher page ⁵²	Provides an influential framework for linking algorithms, representations, and implementation—reshaping what “explanation” in brain science means. ⁵³
1982	John J. Hopfield ⁵⁴ publishes the Hopfield network (emergent collective computation)	PNAS paper ⁵⁵	Establishes attractor-network exemplars that support normal-science puzzle solving in memory, optimization, and dynamics.
1986	David E. Rumelhart ⁵⁶ , Geoffrey Hinton ⁵⁷ , and Ronald J. Williams ⁵⁸ publish backpropagation	Nature record ⁵⁹	Reboots learning-centered network paradigms; later becomes a focal “anomaly” site in neuroscience debates about biological plausibility. ⁵³
1986	California Institute of Technology ⁶⁰ establishes the Computation and Neural Systems PhD program	Caltech history note ⁶¹	Institutionalizes interdisciplinary training—critical for Kuhn’s emphasis on paradigms being reproduced by education and shared exemplars. ²⁸
1987	NeurIPS ⁶² is founded; first proceedings appear for the 1987 meeting	NeurIPS official “founded in 1987” and proceedings archive ⁶³	A key cross-field venue where neural nets and neuroscience interacted; functions as a “trading zone” between communities.
1988	Terrence J. Sejnowski ⁶⁴ , Christof Koch ⁶⁵ , and Patricia S. Churchland ⁶⁶ publish a programmatic <i>Science</i> article “Computational Neuroscience”	Article PDF ²⁷	Explicitly articulates aims, methods, and multi-level challenges, and notes a lack of large-scale theories—key evidence about paradigm status. ⁶⁷

Date	Milestone	Primary / official source(s) to cite	Why this matters for a Kuhnian historiography
1988–1990	Eric L. Schwartz ⁶⁸ edits an early field-defining volume titled <i>Computational Neuroscience</i>	“In press” citations in 1988 + bibliographic record for later publication ⁶⁹	Marks attempts to consolidate exemplars and shared vocabulary via edited volumes—an indicator of paradigm formation. ¹
1988	GENESIS ⁷⁰ begins development; becomes a major neural simulation system	GENESIS reference paper ⁷¹	Software becomes an infrastructural “exemplar” embedding norms about modeling granularity and validation. ⁷²
1989	<i>Neural Computation</i> begins publication (Vol. 1, Issue 1)	MIT Press journal issue page ⁷³	A core publication venue consolidating a community around theory, modeling, and neural computation standards.
1990	Stanley Ogawa ⁷⁴ reports BOLD contrast (early fMRI lineage)	PubMed record ⁷⁵	Enables whole-brain functional measurement; drives new model classes and statistical paradigms oriented to imaging data.
1990	Carver Mead ⁷⁶ publishes “Neuromorphic electronic systems” (neuromorphic engineering)	Proceedings paper PDF	Establishes a hardware-embodied computational paradigm that feeds back into neuroscientific modeling constraints. ⁶⁷
1992	First “CNS Meeting” organized in San Francisco ⁷⁷ (annual computational neuroscience meeting lineage)	Organization for Computational Neurosciences ⁷⁸ history page ⁷⁹	Conferences are key Kuhnian sites where puzzles, exemplars, and standards are stabilized and contested. ¹
1994	NEST Simulator ⁸⁰ first released (as SYNOD), later a major large-network simulator	NEST official site ⁸¹	Supports a normal science of large-scale spiking network modeling with standardized tool assumptions.
1996	Balanced-network / irregular-activity theory becomes prominent (asynchronous irregular states)	PubMed record for van Vreeswijk & Sompolinsky (1996) ⁸²	An exemplar shift in how variability and irregular spiking are interpreted (from “noise” to network regime).

Date	Milestone	Primary / official source(s) to cite	Why this matters for a Kuhnian historiography
1997	NEURON ⁸³ simulation environment described in a canonical methods paper	PubMed record ⁸⁴	Codifies mechanistic modeling workflows and validation practices, embedding paradigmatic assumptions into software.
1999	Predictive coding formalization in visual cortex modeling	Rao & Ballard paper record ⁸⁵	Helps consolidate predictive-processing paradigms that reframe “explanation” as inference under generative models. ⁸⁶
2001–2002	Major textbooks consolidate field training	Theoretical Neuroscience ⁸⁷ and Spiking Neuron Models ⁸⁸ publisher records ⁸⁹	Textbooks are Kuhn’s paradigm carriers: they transmit exemplars and standards to new practitioners. ²⁸
2003	Allen Institute ⁹⁰ founded (large-scale, open data and tools)	Allen Institute “founded in 2003” ⁹¹	Institutional driver of data-centric paradigms and benchmark datasets; enables tool-and-standard reorientations.
2004	COSYNE ⁹² begins (systems + computation community)	COSYNE official site ⁹³	Another key epistemic venue emphasizing tight coupling of theory and data in systems neuroscience.
2005	Optogenetics milestone (millisecond optical control of neural activity)	Boyden et al. 2005 PubMed ⁹¹	Adds causal manipulation at scale, shifting what counts as explanatory adequacy for models (predict → intervene). ⁵³
2005	International Neuroinformatics Coordinating Facility ⁹⁴ founded (neuroinformatics coordination)	INCF history page ⁹⁵	Infrastructure for standards and interoperability; enables cross-lab normal science built on shared data formats.
2005–2024	Blue Brain Project ⁹⁶ runs as a simulation neuroscience initiative	EPFL statement (2005–end of 2024) ⁸²	A prominent “digital reconstruction” research program: tests whether detailed simulation can serve as a central paradigm.

Date	Milestone	Primary / official source(s) to cite	Why this matters for a Kuhnian historiography
2013–2023	Human Brain Project ⁹⁷ runs as EU FET Flagship, later transitioning to EBRAINS infrastructure	HBP official timeline and end notice ⁹⁸	A large-scale institutional attempt to reshape the field via shared infrastructure; offers a case study for Kuhn + sociology-of-science analysis.
2013–present	BRAIN Initiative ⁹⁹ launches in the U.S. (technology and sharing emphasis)	NIH/White House archival pages ¹⁰⁰	Institutional and technological acceleration, aligning with Kuhn's note that “outside the sciences” conditions can affect alternatives and reform. ¹
2014–present	Data standardization movement: Neurodata Without Borders ¹⁰¹ launched mid-2014; pilots yield common formats	NWB origins + pilot paper ¹⁰²	A shift in disciplinary matrix: shared data languages become central exemplars enabling cumulative, multi-site normal science. ³⁷
2017	Neuropixels high-density silicon probes published; major scale-up in cellular recordings	Nature paper ¹⁰³	Reconfigures feasible scientific puzzles (population dynamics at scale), driving new statistical/ML paradigms. ¹⁰⁴
2017–present	BICCN / BICAN cell atlas programs broaden large-scale multimodal atlasing	NIH program pages ¹⁰⁵	Reinforces data-driven paradigms and pressures standardization and integrative modeling across modalities. ³⁷

Mermaid timeline of selected milestones

The following diagram is intended as a *narrative spine* rather than a complete chronology; the table above provides the citable backbone. ¹⁰⁶

timeline

title Computational Neuroscience – Selected Milestones (1907–2024)

1907 : Lapique integrate-and-fire neuron model

1943 : McCulloch & Pitts logical calculus for neural activity

1948 : Wiener publishes Cybernetics (feedback/control/communication)

1949 : Hebb proposes synaptic learning theory (Organization of Behavior)

1952 : Hodgkin-Huxley conductance-based action potential model

1961 : Efficient coding lineage (Barlow, Sensory Communication)
 1964 : Rall dendritic trees and compartmental modeling foundations
 1976 : Patch clamp enables single-channel electrophysiology
 1982 : Marr's Vision (computational levels of analysis)
 1982 : Hopfield networks (collective computation and attractors)
 1986 : Backpropagation popularized; Caltech CNS PhD program founded
 1987 : NeurIPS founded (neural computation + ML venue)
 1988 : Sejnowski/Koch/Churchland programmatic synthesis in Science
 1990 : BOLD contrast (fMRI lineage); neuromorphic systems (Mead)
 1992 : First CNS Meeting (annual computational neuroscience meeting)
 1994 : NEST released (large-scale spiking simulation tool)
 1996 : Balanced-network theory (irregular/asynchronous regimes)
 1997 : NEURON simulation environment canonical description
 1999 : Predictive coding (Rao & Ballard) in cortical modeling
 2001 : Theoretical Neuroscience textbook consolidates training
 2003 : Allen Institute founded
 2004 : COSYNE founded (systems + computation)
 2005 : Optogenetics milestone; INCF founded; Blue Brain launches
 2013 : BRAIN Initiative launches; Human Brain Project begins
 2014 : NWB standardization initiative launched
 2017 : Neuropixels high-density recording probes
 2023 : Human Brain Project formal end; EBRAINS infrastructure continues
 2024 : Blue Brain Project concludes

Candidate Paradigms and Kuhnian Episodes in Computational Neuroscience

Why “candidate paradigms” rather than a single paradigm

SSR ties mature scientific development to acquiring a paradigm that constrains fundamentals and makes normal science possible; it also allows that paradigm change can occur within subspecialties. ¹ Computational neuroscience has repeatedly operated as a boundary domain spanning neuroscience, mathematics, physics, computer science, and engineering, and early syntheses explicitly described it as lacking “successful large-scale theories” and as defined by problems and methods. ¹⁰⁷ This strongly suggests a **multi-paradigm** or partially “pre-paradigmatic” structure at the full-field level, even while many subareas display stable exemplars and normal-science puzzle solving. ¹⁰⁸

Table comparing candidate paradigms

The table uses Kuhn’s “disciplinary matrix” components—symbolic generalizations, models, values, exemplars—to specify what a “paradigm” would mean concretely in computational neuroscience. ¹⁷

Candidate paradigm	Core assumptions	Typical methods	Core problems treated as legitimate	Canonical exemplars (papers/tools/projects)
Conductance-based mechanistic modeling and simulation	Neurons/circuits are explainable via biophysical mechanisms (channels, conductances, morphology) and lawful dynamics	Differential-equation models; compartmental modeling; parameter fitting; simulation environments	Action potential generation; dendritic integration; synaptic mechanisms; micro-to-meso causal explanation	HH model (1952) ⁴² ; Rall dendritic theory (1964) ¹⁰⁹ ; NEURON simulator ⁸⁴ ; GENESIS simulator ⁷¹
Network dynamics and attractor computation	Collective circuit dynamics yield computation; emergent properties can be central explanatory targets	Recurrent networks; dynamical systems; mean-field; regime analysis	Memory as attractor dynamics; pattern completion; stability/chaos; population regimes	Hopfield (1982) ⁵⁵ ; balanced-network theory (1996) ⁸² ; spiking network simulators (e.g., NEST) ⁸¹
Neural coding and information-theoretic representation	Neural activity encodes information; efficiency constraints shape codes	Information theory; encoding/decoding models; tuning curves; dimensionality reduction	What is represented; how reliably; what code is efficient under constraints	Barlow (1961) ⁴⁵ ; multi-unit recording and statistical analysis emphasized in foundational synthesis ²⁷
Normative probabilistic inference and predictive processing	Perception/action as Bayesian (or approximate Bayesian) inference; prediction errors drive updating	Generative models; Bayesian decision theory; predictive coding; variational methods	Explaining perception under uncertainty; integrating cues; prediction error dynamics	Knill & Pouget (2004) ¹¹⁰ ; Rao & Ballard (1999) ⁸⁵ ; Friston free-energy principle (2010) ¹¹¹

Candidate paradigm	Core assumptions	Typical methods	Core problems treated as legitimate	Canonical exemplars (papers/tools/projects)
Data-intensive statistical and ML neuroscience	High-dimensional data require statistical/ML structure discovery; models are judged by predictive adequacy on rich datasets	GLMs; latent-variable models; ML pipelines; standardized data formats; benchmark-driven model comparison	Population computation at scale; cross-session generalization; linking neural manifolds to behavior	Statistical practice guidance (2005) ¹⁰⁴ ; Neuropixels (2017) ¹¹² ; NWB standards (2014–) ¹⁰² ; Allen Observatory datasets ¹¹³
Large-scale “digital brain” reconstruction / simulation neuroscience	Detailed reconstructions and simulations are a complementary pathway to understanding brain structure–function	Multi-scale modeling; HPC simulation; reconstruction pipelines; curated morphological/ connectomic datasets	Whether detailed simulations generate explanatory constraints; integration across scales	Blue Brain Project (2005–2024) ⁸² ; Human Brain Project (2013–2023) ⁹⁸

Episodes of normal science, anomalies, crises, and shifts

In Kuhn’s terms, normal science proceeds under shared exemplars and standards, with anomalies typically treated as puzzles to be resolved rather than as refutations. ¹¹⁴ Within conductance-based mechanistic modeling, the HH model and dendritic compartmental theory functioned as long-lived exemplars that anchored decades of work in parameterizing channels, morphology, and synaptic mechanisms (often reinforced by new experimental access such as patch clamp). ¹¹⁵

Candidate **anomalies** in this mechanistic paradigm are not primarily “failed predictions,” but scaling and explanatory-scope problems: how to responsibly generalize from detailed single-neuron models to network computation and behavior, and how to manage degrees of freedom without losing explanatory traction. A philosophical marker of this methodological tension is the debate about what counts as explanation in computational neuroscience (mechanistic vs merely descriptive or predictive models). ¹¹⁶ Interpreted Kuhnianly, this is pressure on the disciplinary matrix’s *values* (simplicity, realism, explanatory adequacy) and on exemplars of “good modeling.” ³⁷

Several **candidate shifts** look Kuhnian at the subfield level:

First, the conceptual move from treating variability as nuisance to treating irregular activity regimes as characteristic circuit states functions as an exemplar shift in network theory (e.g., balanced/asynchronous regimes). ¹¹⁷ Second, the arrival of high-throughput and multimodal recording technologies (fMRI,

optogenetics, Neuropixels) changed the feasible puzzle set and encouraged statistical and ML paradigms oriented toward prediction and latent structure rather than toward hand-crafted mechanistic detail alone. ¹¹⁸ Third, the rise of standardization infrastructures (INCF; NWB; large atlas networks) reconfigures what Kuhn would call the “disciplinary matrix” by making shared data languages and interoperable toolchains part of normal science itself. ¹¹⁹

Whether these count as Kuhnian **crises** is less clear. SSR stresses that crises arise when anomalies “subvert the existing tradition of scientific practice” and produce extraordinary investigations culminating in new commitments. ¹ In computational neuroscience, major discontinuities often look more like **technology-enabled expansion** and **pluralistic recombination** than wholesale abandonment of an older paradigm; multiple paradigms persist and are frequently integrated (e.g., mechanistic constraints + statistical modeling + normative objectives). ¹²⁰ That pattern pushes historiography toward either (i) a “many micro-revolutions in subspecialties” reading explicitly allowed by Kuhn, or (ii) a non-Kuhnian account centered on research programmes, infrastructures, and trading zones. ¹²¹

Sociological, Institutional, and Technological Drivers of Change

Kuhn explicitly notes that conditions “outside the sciences” can shape which alternatives are available and which reforms are viable, adding an “analytic dimension” important for understanding scientific advance.

¹ In computational neuroscience, four drivers recur: (i) interdisciplinary training institutions, (ii) conferences/journals that consolidate norms, (iii) funding regimes and big-science initiatives, and (iv) enabling technologies and data infrastructures.

Interdisciplinary education is central in SSR because paradigms are reproduced by training in concrete model problems. ²⁸ Computational neuroscience’s institutionalization includes early dedicated programs such as Caltech’s CNS PhD (1986) and later specialized centers like the Gatsby Computational Neuroscience Unit ¹²² (established 1998). ¹²³ These sites function as Kuhnian “paradigm engines,” stabilizing exemplars (courses, codebases) and selecting for shared values about modeling practice. ¹⁷

Conferences and journals provide the social machinery of normal science: they define what counts as a good puzzle, a satisfying solution, and legitimate standards of evaluation. ¹ Key examples include the CNS meeting lineage and the formation of Organization for Computational Neurosciences ⁷⁸ (in California, 2004) as well as COSYNE (emphasizing empirical-theoretical systems neuroscience exchange). ¹²⁴ The journal *Neural Computation* (Vol. 1 in 1989) served as an additional consolidation mechanism for theory/modeling/statistics communities. ¹²⁵

Funding and institutional initiatives increasingly shape the field’s “disciplinary matrix” by prioritizing certain data types, scales, and infrastructures. Examples include the National Science Foundation ¹²⁶’s Collaborative Research in Computational Neuroscience (CRCNS) program (explicitly encouraging state-of-the-art computational tools across levels) and the NIH-led BRAIN Initiative (technology development, sharing, and large-scale data generation), along with EU-scale programs like the Human Brain Project. ¹²⁷

Technological and infrastructural shifts are unusually central to computational neuroscience: patch clamp (single channels), fMRI (whole-brain dynamics), optogenetics (causal intervention), and Neuropixels (large-scale cellular populations) redefine feasible puzzles and acceptable models. ¹²⁸ Data standards such as NWB and coordinating bodies like INCF convert what might once have been “local craft” into community-wide normal science built on shared formats and tools. ¹²⁹

Mermaid entity-relationship diagram of key actors and institutions

This diagram encodes a **minimal** actor/institution network sufficient for historiographic analysis of paradigm formation (training, venues, infrastructure, funding). It is illustrative rather than exhaustive. 130

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erDiagram
    PERSON ||--o{ PROGRAM : "trains_in"
    PROGRAM }o--|| INSTITUTION : "hosted_by"
    PERSON ||--o{ TOOL : "develops_or_popularizes"
    PERSON ||--o{ PAPER : "authors"
    PERSON ||--o{ PROJECT : "leads_or_shapes"
    PROJECT }o--|| INSTITUTION : "based_at"
    CONFERENCE }o--|| COMMUNITY : "convenes"
    JOURNAL }o--|| COMMUNITY : "publishes_for"
    FUNDER ||--o{ PROJECT : "funds"
    FUNDER ||--o{ PROGRAM : "funds"
    STANDARD ||--o{ COMMUNITY : "enables_interop"
    TOOL ||--o{ COMMUNITY : "embeds_practices"

    PERSON {
        string name
    }
    INSTITUTION {
        string name
    }
    PROGRAM {
        string name
    }
    CONFERENCE {
        string name
    }
    JOURNAL {
        string name
    }
    PROJECT {
        string name
    }
    FUNDER {
        string name
    }
    STANDARD {
        string name
    }
    TOOL {
        string name
    }
    PAPER {
```

```

    string name
}
COMMUNITY {
    string name
}

%% Instances (labels only; not exhaustive)
PERSON ||--o{ PAPER : "writes"
PERSON ||--o{ PROJECT : "directs"

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Kuhn's Fit, Limits, and Alternative Historiographic Frameworks

Where Kuhn fits well

Kuhn's framework is highly productive for computational neuroscience when applied to well-bounded subcommunities where paradigms can be concretely identified as exemplars plus training and standards.

¹³¹ For example, the conductance-based modeling tradition has clear canonical exemplars (HH; dendritic compartmental theory), tool-embedded practices (NEURON/GENESIS), and validation norms shaped by electrophysiological measurement. ¹³² Similarly, data-intensive computational neuroscience around high-density recording has recognizable infrastructural exemplars (Neuropixels), data standards (NWB), and statistical modeling norms codified in methodological syntheses. ¹³³

Kuhn's emphasis on **education and textbooks** as paradigm transmitters maps cleanly onto computational neuroscience, where textbooks and software tutorials often function as the most powerful vehicles for exemplar transmission. ¹³⁴ His detailed treatment of translation and "going native" helps explain why cross-paradigm disputes in the field so often hinge on learning a new formal language (e.g., probabilistic modeling, deep nets, or new data-languages) rather than on pointwise empirical "refutation." ¹³⁵

Where Kuhn fits poorly or requires modification

At the level of "the whole field," computational neuroscience has frequently lacked the Kuhnian signature of a single dominant paradigm that suppresses fundamental controversy. Kuhn himself states that acquiring a paradigm is a sign of a field's maturity; early programmatic accounts in computational neuroscience, by contrast, explicitly described the domain as lacking large-scale theories and being defined by problems and methods. ¹³⁶ This suggests either (i) a **pre-paradigm** or **multi-paradigm** condition (consistent with Masterman's observation of paradigm polysemy), or (ii) a field whose primary organizing structures are infrastructural and methodological rather than theoretical. ¹³⁷

Additionally, computational neuroscience's major discontinuities are often **technology- and infrastructure-driven** (recording, imaging, intervention, standardization), whereas SSR's canonical story emphasizes anomaly-to-crisis dynamics within relatively stable observational regimes. ¹³⁸ In a domain where new instruments change what can be observed and manipulated, "revolution" may be precipitated less by anomaly accumulation and more by shifts in feasible experimental/computational coupling, funding priorities, and data-sharing infrastructure. ¹³⁹

Alternative and complementary frameworks

Because computational neuroscience often hosts **coexisting** approaches (mechanistic, normative, statistical/ML, simulation-first), Imre Lakatos ⁷’ framework is a strong complement: it models science as competing research programmes with “hard cores” and protective belts, and it provides criteria for progressive versus degenerative problemshifts without requiring a single paradigm monopoly. ¹⁴⁰ Larry Laudan ⁸’s research traditions similarly accommodate persistent pluralism and problem-solving success as the unit of progress rather than paradigmatic replacement.

For interdisciplinary coordination, Peter Galison ⁹’s “trading zones”—sites where communities with different languages and standards coordinate via partial languages, tools, and local agreements—maps closely to computational neuroscience’s cross-field conferences, shared software ecosystems, and standardized data formats. ¹⁴¹ Ian Hacking ¹⁰’s “styles of reasoning” and Pierre Bourdieu ¹¹’s field theory add sociological depth for understanding why some modeling styles become dominant (institutional capital, funding, prestige structures) even when alternative styles remain viable.

Finally, Kuhn’s own later work emphasizes more fine-grained, taxonomic/linguistic aspects of incommensurability; this can sharpen historiography of computational neuroscience where disputes often concern redefinition of key terms (e.g., “representation,” “computation,” “prediction error,” “mechanism”) across subcommunities. ¹⁴²

Primary Sources to Prioritize and an Archival Strategy

A Kuhnian historiography gains credibility by building its narrative around **primary exemplars** (papers, software, benchmark datasets) and **official community artifacts** (programs, proceedings, standards documents), because these are the loci of “disciplinary matrix” commitments and exemplar transmission.

¹⁷

Primary sources to prioritize include foundational model papers (Lapicque; HH; Rall; Hopfield; predictive coding; free-energy principle), instrumentation milestones (patch clamp; BOLD/fMRI; optogenetics; Neuropixels), and institutional consolidations (early programmatic syntheses and textbooks). ¹⁴³

Official sources to prioritize for documenting community formation and epistemic norms include conference archives and society histories (OCNS/CNS meetings; COSYNE; NeurIPS proceedings), journal archives (*Neural Computation*), and infrastructure/standards documentation (INCF, NWB), alongside major initiative portals (BRAIN Initiative; Human Brain Project/EBRAINS; Blue Brain). ¹⁴⁴

To anchor “major labs/centers” without drifting into anecdote, prioritize institutions whose public documentation explicitly frames computational neuroscience as a core identity and provides stable publication and tool outputs. Suitable starting points include the Salk Institute for Biological Studies ¹⁴⁵’s computational neurobiology presence (e.g., Sejnowski Lab), the Gatsby Unit (theory + learning at scale), and open-science neurotechnology hubs such as Janelia Research Campus ¹⁴⁶’s Neuropixels ecosystem (developed via collaborations including Howard Hughes Medical Institute ¹⁴⁷ and the Allen Institute). ¹⁴⁸

A practical archival workflow is: (1) reconstruct shifting “exemplars” via the timeline table’s cited papers and tools; (2) map community consolidation through conference programs, proceedings, and textbook editions;

(3) document infrastructure and funding shifts through official initiative pages and standards documents; and (4) only then use secondary historiography (e.g., philosophical analyses of explanation in computational neuroscience) to interpret how explanatory norms and field values changed. ¹⁴⁹

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